

Project Initialization and Planning Phase

Date	28 JUNE 2026
Team ID	XXXXXX
Project Title	EduGenie Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

EduGenie is an AI-powered learning assistant that helps students learn more effectively by providing instant answers, detailed explanations, concise summaries, quizzes, and personalized learning paths. The system uses Google's Gemini AI model to deliver accurate and interactive educational support through a simple web interface.

Project Overview	
Objective	To develop an AI-powered educational assistant that helps students understand concepts, answer academic questions, summarize study material, generate quizzes, and provide personalized learning paths.
Scope	The project allows users to: • Ask academic questions and receive AI-generated answers. • Get detailed explanations of difficult concepts. • Summarize lengthy paragraphs. • Generate quizzes for self-assessment. • Create personalized learning paths for any subject.
Problem Statement	
Description	Many students spend a significant amount of time searching different websites for explanations, notes, quizzes, and study plans. This makes learning inefficient and time-consuming.
Impact	Providing all these educational features in one AI-powered application improves learning efficiency, saves time, increases understanding, and enhances exam preparation.
Proposed Solution	
Approach	Develop an AI-powered web application using Python, FastAPI, HTML, CSS, JavaScript, and Google's Gemini API to provide intelligent educational assistance through multiple learning features.
Key Features	• AI-powered Question Answering • Concept Explanation Generator • Paragraph Summarizer • Quiz Generator • Personalized Learning Path Generator • Simple and User-friendly Interface • Fast API-based backend • Gemini AI integration

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU	Intel Core i5 Processor
Memory	RAM	8 GB
Storage	256 GB	SSD
Software		
Frameworks	Backend Framework	FastAPI
Libraries	Additional libraries	google-generativeai, python-dotenv, uvicorn, requests
Programming Language	Development Language	Python 3.12
AI Service	Large Language Model	Google Gemini API
Frontend	User Interface	HTML, CSS, JavaScript
Data		
Data	Source, size, format	User questions, study topics, paragraphs, quiz topics, learning path requests (processed through Gemini API)